Example 7.59

Find the local extremum of the function $f(x) = x^4 + 32x$.

Solution

We have,

$$f'(x) = 4x^3 + 32 = 0 \text{ gives } x^3 = -8$$

$$\Rightarrow x = -2$$

and $f''(x) = 12x^2$.

As $f''(-2) > 0$, the function has local minimum at $x = -2$. The local minimum value is $f(-2) = -48$. Therefore, the extreme point is $(-2, -48)$. ■

Example 7.60

Find the local extrema of the function $f(x) = 4x^6 - 6x^4$.

Solution

Differentiating with respect to x, we get

$$f'(x) = 24x^5 - 24x^3$$

$$= 24x^3(x^2 - 1)$$

$$= 24x^3(x+1)(x-1)$$

$f'(x) = 0 \Rightarrow x = -1, 0, 1$. Hence the critical numbers are $x = -1, 0, 1$.

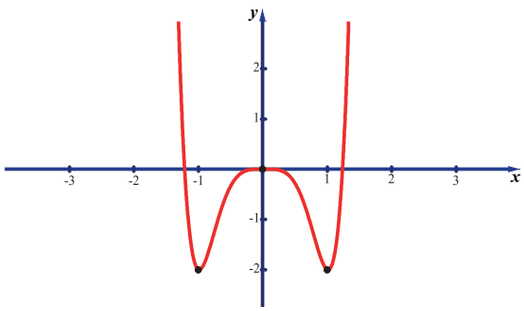


Fig.7.26

Now, $f''(x) = 120x^4 - 72x^2 = 24x^2(5x^2 - 3)$.

$\Rightarrow f''(-1) = 48$, $f''(0) = 0$, $f''(1) = 48$.

As $f''(-1)$ and $f''(1)$ are positive by the second derivative test, the function $f(x)$ has local minimum. But at $x = 0$, $f''(0) = 0$. That is the second derivative test does not give any information about local extrema at $x = 0$. Therefore, we need to go back to the first derivative test. The intervals of monotonicity is tabulated in Table 7.8.

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
Sign of $f'(x)$	$-$	$+$	$-$	$+$
Monotonicity	strictly decreasing	strictly increasing	strictly decreasing	strictly increasing

Table 7.8

By the first derivative test $f(x)$ has local minimum at $x = -1$, its local minimum value is -2. At $x = 0$, the function $f(x)$ has local maximum at $x = 0$, and its local maximum value is 0. At $x = 1$, the function $f(x)$ has local minimum at $x = 1$, and its local minimum value is -2. ■

Remark

When the second derivative vanishes, we have no information about extrema. We have used the first derivative test to find out the extrema of the function!

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